

Padārtha: Ontology-Grounded Fine-Grained NER Benchmark for Classical Sanskrit

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Abstract

Annotation schemas are not neutral. When applied to classical literature, tag sets developed for modern journalistic texts impose source-culture definitions on texts they were never designed to describe. We instead ground a schema in the tradition of the text itself introducing *Padārtha*, the first ontology-grounded fine-grained Named Entity Recognition (NER) benchmark for Sanskrit, built on the *Mahābhārata* epic. Our tag set derives from *Nyāya-Vaiśeṣika*, a classical Indian ontological system, yielding 18 fine-grained categories organized under 10 ontological nodes and mapped onto five standard coarse tags, ensuring interoperability with existing benchmarks. Expert annotators label over 12.6K entries from a scholarly index of named entities, linked to corresponding mentions in the *Mahānāma* corpus, producing fine-grained annotations for 108,335 entity mentions across 73,632 verses, along with a 5,000-verse expert-verified test set sampled to stress rare mentions. We present the first systematic benchmarking of generative NER against traditional architectures for Sanskrit, finding that fine-tuned generative models perform comparably to task-specific systems. However, all systems show a sharp decline from coarse to fine granularity and struggle with out-of-entity mentions unseen during training. The limitation is not due to data scarcity alone, as fine-tuned models recall unseen entities far worse than seen ones and tend to default to the majority sense under lexical ambiguity.

1 Introduction

As a core task in natural language understanding, Named Entity Recognition (NER) is widely adopted across domains such as news (Tedeschi and Navigli, 2022), medical texts (Luo et al., 2022), and social media (Derczynski et al., 2017). Despite its broad applicability, current systems perform well on coarse categories but struggle with nuanced

or specialised entity types (Ding et al., 2021; Katz et al., 2023; Fetahu et al., 2023), with performance dropping further as categories move deeper into a hierarchy (Katz et al., 2023). These challenges have been studied almost exclusively in standard domains; literary text, where entities follow distinct distributions and narrative behaviour, remains comparatively neglected (Vala et al., 2015; Bamman et al., 2019; Silva and Moro, 2024; Zhao et al., 2025).

Creating fine-grained NER datasets for literary texts, particularly classical corpora such as epics, religious works, and historical literature, is challenging because standard tagsets either lack sufficient granularity or are designed for modern, journalistic domains with limited domain-specific types (Liu et al., 2021). These limitations are amplified in texts containing culturally specific items (CSIs). Schema like OntoNotes do not represent mythological entities (Tedeschi and Navigli, 2022), while CoNLL-style tagsets remain overly coarse, relying on ill-defined catch-all categories such as MISC (Mayhew et al., 2024). The difficulty is not that a label is missing but that the underlying partition of the world is wrong: “Garuḍa”, for instance, is simultaneously a bird, a deity, the son of Kaśyapa and Vinatā, and the vehicle of Viṣṇu, and HiNER (Murthy et al., 2022) reports precisely this class of entity as resistant to assignment under PERSON. Consequently, if such schemes are not applied with due care, they risk overlooking the complexity of the target culture and imposing source-culture concepts onto it (Hershcovich et al., 2022).

Beyond schema limitations, annotated literary data are scarce (Zhao et al., 2025), especially for historical texts where corpus construction requires costly expert knowledge (Palladino, 2026). Unlike standardized news text, fiction exhibits substantial lexical variation with diverse, context-dependent naming conventions (Han et al., 2021), and a few major entities typically dominate literary dis-

course (Manikantan et al., 2024). This skewed distribution leaves rare entities highly susceptible to out-of-entity (OOE) failures (Jiang et al., 2025), where mention tokens at test time are entirely unseen during training.

This raises a question that the literary NER literature has not addressed: when a text’s entities resist an imported tagset, is it better to extend that tagset ad hoc, or to ground the schema in a categorical system internal to the text’s own intellectual tradition? We argue for the latter. An *emic* schema, one drawn from within the tradition, fixes category boundaries that were articulated independently of the annotation task, and so cannot be accused of being drawn to fit the data post hoc; this is the constructive counterpart to the warning of Hershcovich et al. (2022).

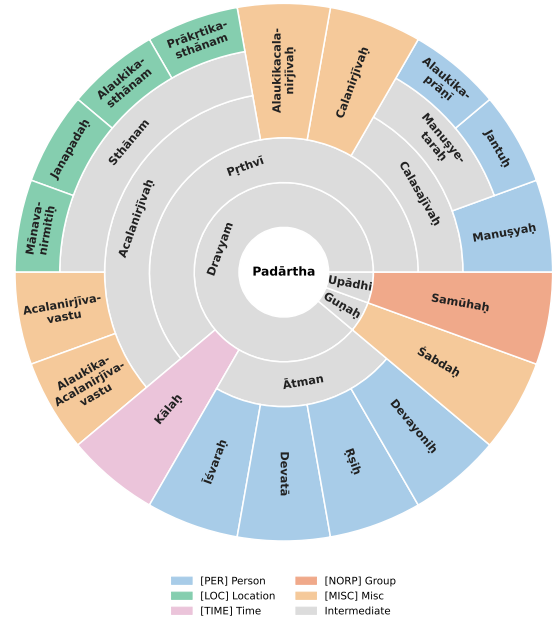
To bridge these gaps and investigate the unique challenges of the literary domain, we introduce *Padārtha*, the first fine-grained NER benchmark in Sanskrit, based on the *Mahābhārata*, one of the longest epics in world literature. Its vast narrative captures diverse naming conventions, while its composition in Sanskrit, a morphologically rich language, presents additional computational hurdles and establishes a typological contrast to standard English benchmarks. In this work, our core contributions are:

(1) Ontology-Grounded Hierarchical Tagset

We propose an annotation scheme grounded in Sanskrit linguistic extensions of Nyāya–Vaiśeṣika (NV) (Nair et al., 2013), an ancient Indian ontological framework, to capture culturally specific items (CSIs) at the granularity the epic demands. The scheme has 18 fine-grained leaves under 10 ontology nodes, each mapping to one of five standard tags (PER, LOC, NORP, MISC, TIME), so it stays compatible with existing NER benchmarks (Figure 1).

(2) Fine-Grained Sanskrit NER Dataset We build *Padārtha* based on the *Mahānāma* corpus (Sarkar et al., 2025), which pre-links mentions to Sørensen’s *Index* (Sørensen, 1904). A Sanskrit scholar labeled 12.6K index entries from their descriptions using our schema; projecting these labels through the existing links gives 73,632 verses and 108,335 mentions. We then read and corrected a 5,000-verse test set by hand to serve as gold evaluation data.

(3) Comprehensive Benchmarking Motivated by the shift toward modern LLMs, we benchmark



padārtha = ontological root; *dravya* = substance; *guṇa* = quality; *upādhi* = incidental property; *prthvī* = earth; *ātman* = soul; *sthānam* = places; *calasajīvaḥ* = mobile living beings; *acalanirjīvaḥ* = immobile non-living things; *manuṣyetaṛaḥ* = non-human beings.

Figure 1: Nyāya-Vaiśeṣika ontology hierarchy underlying the tagset. Leaf nodes correspond to fine-grained categories (glosses and examples in Table 1). Color indicates standard coarse category (legend).

decoder-only LLMs under both parameter-efficient fine-tuning and few-shot in-context learning, contextualized against a CRF baseline, a fine-tuned encoder, and a generative encoder-decoder. Performance degrades consistently as categorical granularity increases, evaluated on a test split constructed to capture rare and unseen OOE mentions.

(4) Empirical and Theoretical Analysis of Fine-Grained Challenges

We analyze the degradation at finer granularity along four dimensions: memorization vs. generalization, lexical ambiguity, information-theoretic difficulty, and qualitative error patterns. Fine-tuned models show substantially lower recall on unseen entities and default to the majority sense under ambiguity. Information-theoretic analysis confirms that context, rather than the entity itself, serves as the primary signal.

2 Tagset Design and Dataset Construction

2.1 The Nyāya-Vaiśeṣika Ontology and NER Adaptation

The Nyāya Vaiśeṣika (NV) school unifies Nyāya, founded by Gautama (c. 300 BCE) and centered on

SL	Coarse	Fine Tag	Gloss	Example	Sørensen Description
1	Person	<i>īśvaraḥ</i>	The Trimūrti (Trinity)	Śiva	Great god (Mahādeva)
2	Person	<i>devatā</i>	Gods	Indra	King of the gods
3	Person	<i>ṛṣiḥ</i>	Sages	Vasiṣṭha	Celebrated celestial sage
4	Person	<i>devayoniḥ</i>	Divine-origin beings	Citraratha	King of the Gandharvas
5	Person	<i>manuṣyaḥ</i>	Humans	Arjuna	Third Pāṇḍava prince
6	Person	<i>jantuh</i>	Animals	Sugrīva	One of Kṛṣṇa's horses
7	Person	<i>alaukikapraṇī</i>	Mythical creatures	Hamsikā	A celestial cow
8	Location	<i>prākṛtikasthānam</i>	Natural locations	Daṇḍakāraṇya	A forest
9	Location	<i>alaukikasthānam</i>	Mythical places	Gandharvaloka	World of the Gandharvas
10	Location	<i>janapadaḥ</i>	Geo-political entities	Aṅga	The Aṅga country
11	Location	<i>mānavanirmitaḥ</i>	Man-made structures	Agastyāśrama	Hermitage of Agastya
12	Misc	<i>calanirjivaḥ</i>	Mobile artifacts	Jaitra	A chariot
13	Misc	<i>acalanirjivavastu</i>	Immobile artifacts	Asamprṣṭha	An immobile entity/artifact
14	Misc	<i>alaukikacalanirjivaḥ</i>	Mythical mobile artifacts	Puṣpaka	A celestial car
15	Misc	<i>alaukikacalanirjivavastu</i>	Mythical immobile artifacts	Sudarśana	The discus of Kṛṣṇa
16	Misc	<i>śabdaḥ</i>	Texts and mantras	Nitiśāstra	Science of ethics
17	NORP	<i>samūhaḥ</i>	Nationalities, Religious, Groups	Ābhīra	People west of the Indus
18	Time	<i>kālāḥ</i>	Units of time	Bhādrapada	Name of a month

Table 1: Complete coarse and mapped fine-grained Nyāya-Vaiśeṣika tagset, featuring examples and their canonical descriptions from the book *Index to the Names in the Mahābhārata* (Sørensen, 1904).

epistemology and logic, with Vaiśeṣika, founded by Kaṇāda, which develops a systematic ontology of existence (Nair et al., 2013). It classifies reality into seven *padārthas*, of which *dravya* (substance) is most relevant to Named Entity Recognition. The Vaiśeṣika Sūtra defines *dravya* as *kriyāguṇavat samavāyikāraṇam*, that which possesses actions and attributes and serves as an inherent cause (VS I.1.15) (Sinha, 1923), supporting the treatment of entities as discrete objects with properties and relations.

Our fine-grained tagset draws directly on the extended Vaiśeṣika *dravya* classification of Nair et al. (2013), which refines the classical scheme and adds *alaukika* (celestial) nodes to accommodate entities in the Amarakośa lexicon. Fifteen of our eighteen categories (Table 1) are inherited from this *dravya* hierarchy, preserving category names and scope; the selection process is described in §2.3. To cover entities outside the substance branch, we include *samūhaḥ* and *śabdaḥ* from the *upādhi* (adventitious property) and *guṇa* (quality) nodes, respectively. We further extend the framework with *janapadaḥ* to expand the classification of places (*sthānam*) beyond man-made and natural categories.

Unlike standard flat NER schemas, the NV-derived taxonomy is hierarchical, with each terminal label reached through intermediate categories rather than assigned independently (Figure 1). For

instance, *manuṣyaḥ* (humans) follows substance → earth → mobile living beings → *manuṣyaḥ*, while *prākṛtikasthānam* (natural locations) follows substance → earth → immobile non-living things → places → *prākṛtikasthānam*, sharing a common root before diverging. All eighteen terminal categories map to five coarse NER tags (Table 1), preserving compatibility while retaining hierarchical structure.

2.2 Base Resource: The Mahānāma Corpus

We build *Padārtha* upon the *Mahānāma* corpus (Sarkar et al., 2025), the first large-scale Sanskrit entity linking dataset derived from the Indian epic *Mahābhārata*. The epic's extended narrative poses significant entity resolution challenges due to the high variability and ambiguity of classical names (Sarkar et al., 2025). *Mahānāma* links entity mentions to canonical entities in a knowledge base, where each entity aliases is associated with a description from Sørensen's *Index to the Names in the Mahābhārata* (Sørensen, 1904)¹. These descriptions, covering approximately 12.6K entries, serve as the primary basis for our ontological annotation (Table 1). *Mahānāma* provides three coarse-grained tags (PER, LOC, and MISC); however, it does not provide any guidelines on classification

¹<https://www.sanskrit-lexicon.uni-koeln.de/scans/INMScan/2020/web/index.php>

Format	Used By	Example Output (Mahābhārata CE Volume 1 Chapter 1 Verse 2)
BIO Tags	CRF, MuRIL, ByT5	loma (B-PER) harṣaṇa (I-PER) putra (I-PER), ugraśravāḥ (B-PER), sautiḥ (B-PER), paurāṇikā (B-PER), u (O), naimiṣa (B-LOC), araṇye (O), śaunakasya (B-PER) ...
Inline Bracketed	Qwen2.5-7B, Gemma-4-12B (fine-tuning)	[loma harṣaṇa putra PER] [ugraśravāḥ PER] [sautiḥ PER] [paurāṇikā PER] u [naimiṣa LOC] araṇye [śaunakasya PER] ...
Two-Stage Pipeline	Qwen2.5-7B, Gemma-4-12B, Gemini 3 Flash (few-shot)	1. Mention Detection: @loma harṣaṇa putra## @@ugraśravāḥ## @sautiḥ## @paurāṇikā## u @@naimiṣa## araṇye @@śaunakasya## ... 2. Classification: ["PER", "PER", "PER", "PER", "LOC", "PER", ...]

Table 2: Example verse and output representation formats by model and training paradigm, shown on the same example verse (“... sage Lomaharṣaṇa’s son Ugraśravas [=Sauti], well-read in the Purāṇas, [came to] the Naimiṣa forest, [during] Śaunaka’s sacrifice...”). All formats are converted to a unified BIO representation for scoring. In the Two-Stage Pipeline row, @@...## mark candidate mentions from the detection stage, following the cascade design of (Luo et al., 2025).

and does not conduct any experiments on NER models.

2.3 Tag Set

The extended NV ontology Nair et al. (2013) contains substantially more categories. We narrowed it to eighteen fine-grained tags (Table 1) using two criteria. First, we retained only categories functioning as named entities rather than common nouns, excluding branches such as plants, minerals, the *tejaḥ* branch, and all *guṇaḥ* subdivisions except *śabdaḥ*. Second, we collapsed distinctions with insufficient corpus support into their parent node,² except where collapsing would merge semantically distinct siblings, in which case we retained the sparse node (e.g., *mānavanirmitaḥ*, *calanirjivāḥ*, *alaukikacalanirjivāḥ*).

Our coarse tagset maps to standard NER schemas. We adopt PERSON, LOCATION, and MISC from the CoNLL convention (Tjong Kim Sang and De Meulder, 2003), but omit ORG, since its standard definition, institutional or organizational entities, does not meaningfully apply to classical Sanskrit texts. We additionally introduce two coarse categories from OntoNotes (Hovy et al., 2006), NORP and TIME, to accommodate fine-grained NV categories (*samūhaḥ*, *kālaḥ*) that CoNLL’s scheme would otherwise conflate with unrelated entities under tags such as MISC (§2.2). The resulting five-tag coarse scheme remains a strict refinement

of PER/LOC/MISC and can be trivially collapsed back to it for compatibility with existing tools.

2.4 Annotation

Annotation was performed at the level of Index entries linked to Mahānāma mention spans. A doctoral researcher in Sanskrit assigned one of the eighteen NV derived fine grained tags to each of the 12.6K Index entries based on its description (Table 1), consulting the original verse when needed. The assigned label was then inherited by all linked mentions. To account for potential omissions in Mahānāma’s mention marking (Sarkar et al., 2025), the annotator additionally manually reviewed and corrected a 5,000 verse test set.

Unlike modern texts, classical Sanskrit narrative frequently identifies entities through epithets, derived forms, and relational compounds rather than repeating proper names. Since this is a common way of expressing entity mentions in classical texts, the Index treats such expressions as name references, and we followed the convention: *derived forms* (e.g., *Brāhmī* from *Brahmā*, *Aindrī* from *Indra*), *relational identifiers* (e.g., *Kuntīputra*, "son of Kuntī"), and *epithets* (e.g., *Vṛkodara*-“he of the voracious appetite”-for *Bhīma*) are all classified under the entity they identify.

2.4.1 Annotation Reliability

We assess annotation reliability by comparing the primary annotator’s labels against a postdoctoral Sanskrit grammarian with no prior exposure to the Index of Names, over 1,000 doubly annotated test verses (Table 3). Both granularities show substan-

²E.g., the six habitat-based subdivisions of *jantuḥ* and the *apauruṣeyam/pauruṣeyam* distinction within *śabdaḥ*; the latter mirrors OntoNotes’ single WORK_OF_ART tag, which similarly does not subdivide by origin (Hovy et al., 2006).

Level	Agreement (%)	κ
Coarse label	91.03	0.728
Fine label	79.71	0.746
Span detection	F1 = 0.811	

Table 3: Annotation reliability between the primary annotator and a second reviewer, over 1000 doubly annotated test verses.

tial agreement (Landis and Koch, 1977). Coarse κ is lower than fine κ despite higher raw agreement, a known effect of label-distribution skew on chance-corrected measures.

2.5 Dataset Statistics

The final dataset contains 108,335 entity mentions across eighteen fine-grained categories (Table 5) and five coarse groupings (Table 4). Figure 2 shows the rank–frequency distribution of entity mentions, where both surface forms and their lemmatized counterparts exhibit a long-tailed skew, with surface forms spanning a wider rank range due to morphological variation.

Category	Count	Category	Count
PERSON	88,930	MISC	4,537
NORP	9,416	LOCATION	4,798
TIME	654		
Total	108,335 mentions		

Table 4: Distribution of coarse-grained entity categories.

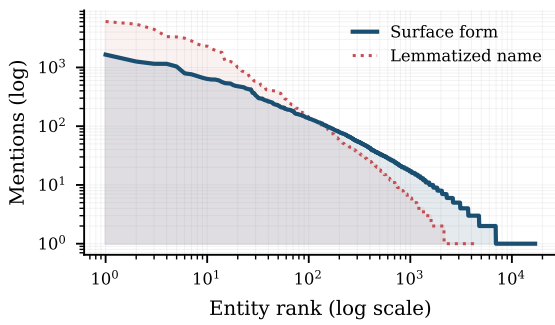


Figure 2: Rank-frequency distribution of entity mentions

2.6 Data Split

From 73,632 verses we selected 5,000-verse test set by sampling verses with rare names (appearing in ≤ 5 verses), ensuring stratified coverage across fine-grained types. Because selection is verse-level, non-rare entities also appear; overall, 73.14%

Tag	Count	Tag	Count
manuṣyaḥ	51,969	devatā	11,149
īśvaraḥ	11,271	samūhaḥ	9,416
ṛṣiḥ	6,424	devayoniḥ	6,022
prākṛtikasthānam	3,284	śabdaḥ	3,132
alaukikapṛāṇī	1,939	alaukika_acalanirjīvavastu	856
alaukikasthānam	806	janapadaḥ	690
kālāḥ	654	acalanirjīvavastu	512
jantuḥ	156	alaukikacalanirjīvaḥ	20
mānavanirmitaḥ	18	calanirjīvaḥ	17
Total	108,335 mentions		

Table 5: Distribution of fine-grained entity categories.

of unique surface forms, accounting for 50.47% of all test mentions, are unseen during training. The train–test distribution across coarse- and fine-grained schemas is shown in Table 10.

3 Experimental Setup

3.1 Models Used

For Sanskrit NER, we consider models spanning four architectural paradigms: a non-neural CRF (Sutton and McCallum, 2012) baseline trained on surface features; MuRIL-large, a multilingual encoder pretrained on Indic languages (Khanuja et al., 2021); ByT5-Sanskrit, an encoder-decoder pretrained on Sanskrit segmentation and morphology (Nehrdich et al., 2024); Qwen2.5-7B-Instruct (Qwen et al., 2025) and Gemma-4-12B (Team et al., 2024b), decoder-only models evaluated under both fine-tuning and few-shot prompting; and Gemini 3 Flash, a closed-source frontier model (Team et al., 2024a) evaluated few-shot only (full settings in Appendix B).

Baseline. We train a linear-chain CRF on shallow surface features (the token itself, character prefixes/suffixes, and a ± 1 token context window), establishing a non-neural baseline.

Encoder Fine-tuning. MuRIL-large is trained using the SpanMarkerNER framework³, following standard span-based BIO sequence labeling (learning rate 1e-5, batch size 8, up to 10 epochs).

Encoder-Decoder Fine-tuning. We use ByT5-Sanskrit, pretrained on Sanskrit segmentation and morphological tagging. To test whether prior training on related Sanskrit tasks transfers to NER, we keep the model’s original token_tag generation format, substituting NER labels (learning rate 3e-4, batch size 8, 15 epochs).

³<https://github.com/tomaarsen/SpanMarkerNER>

Decoder Fine-tuning. Qwen2.5-7B-Instruct and Gemma-4-12B-it are fine-tuned via LoRA (full hyperparameters in Appendix B) using the Inline Bracketed output format (Table 2; full templates in Appendix C, Figure 4), which Zhan et al. (2026) found to outperform alternative output formats for LLM-based NER.

In-Context Learning. Following CascadeNER’s two-stage strategy (Luo et al., 2025), we adopt a mention-detection-then-classification pipeline for Qwen2.5-7B-Instruct, Gemma-4-12B-it, and Gemini 3 Flash. Stage 1 marks candidate spans inline within the full sentence; Stage 2 classifies each span as a JSON label list, with 5-shot demonstrations selected via k -nearest-neighbor retrieval using ByT5-Sanskrit encoder embeddings (full prompts in Appendix C, Figure 5).

Script. ByT5-Sanskrit uses IAST transliteration⁴, consistent with its original pretraining; all other models use Devanagari.

Prompt Design Fine-tuning uses a single instruction specifying the full tagset (gloss and examples per tag), span-tagging rules, and strict output formatting. Few-shot instead uses the two-stage cascade described above, with an added UNKNOWN label in Stage 2 to reject spans Stage 1 over-generates. Both paradigms share the same entity-span rules (proper names, derived names, relational identifiers, epithets). Full prompts in Appendix C.

3.2 Evaluation Metrics

All model outputs are converted to a unified BIO token-tag representation and evaluated using seqeval⁵ with the IOB2 scheme in strict (CoNLL-style) mode, requiring exact span boundary and label match. We report Precision, Recall, and F1, at both coarse and fine granularity. Given severe class imbalance (§2.5), we report both Micro-F1, dominated by frequent categories, and Macro-F1, which weights all categories equally and better reflects performance on sparse tags.

4 Results

4.1 Overall Performance

Fine-tuned neural models substantially outperform the CRF baseline (Table 6), whereas few-shot

decoder-only models (Qwen-FS, Gemma-FS) underperform it at both granularities. Fine-tuning consistently improves over few-shot for the same backbone. Compared to MuRIL, gains are mixed: MuRIL retains the highest Macro-F1, while only Qwen-FT narrowly leads on Micro-F1 (per-label breakdown in Appendix D.1).

Mention detection performance (Table 7) confirms that the primary recall bottleneck across models stems fundamentally from span extraction failures driven by the Out of Entity challenge (§1, §4.2). Gemini 3 Flash is the exception, with recall exceeding precision. All models degrade from coarse to fine prediction, more in Macro F1 (avg. −13.8pp) than Micro F1 (avg. −9.0pp), reflecting low support categories and increased disambiguation demands.

Gemini 3 Flash achieves the highest Micro F1 at both granularities, surpassing all fine tuned models, while MuRIL retains the best Macro F1. Given the Mahābhārata’s public availability, some pretraining exposure is possible, though our ontology differs from existing annotations.

4.2 Seen vs. Unseen Entity Recall

We define seen and unseen based on exact surface-form overlap between training and test data. All models recall seen entities far better than unseen ones, confirming the Out-of-Entity challenge (full results in Table 8). CRF and fine-tuned models show the largest seen/unseen disparities, indicating heavy reliance on memorization, whereas Gemini retains substantially higher unseen recall (55.6–67.2%).

4.3 Impact of Lexical Ambiguity

Mahānāma paper (Sarkar et al., 2025) identifies contextual ambiguity as a central challenge, where identical expressions refer to different entities depending on context. To assess its impact on NER, we analyze performance on minority-sense homonyms, where the correct fine-grained tag contradicts the most frequent surface-form label (Appendix F). We found 1227 minority sense mentions in test data. Fine-tuned models (MuRIL, Gemma-FT, Qwen-FT) consistently fall into a majority-sense trap, with wrong-label rates (27.4%–32.6%) matching or exceeding correct predictions.

4.4 V-information Analysis

Following Ma et al. (2023), we train MuRIL-based entity-only and context-only classifiers to compute

⁴https://en.wikipedia.org/wiki/International_Alphabet_of_Sanskrit_Transliteration

⁵<https://pypi.org/project/seqeval/>

Methodology	Architecture	Model	Output Format	Micro Average (%)			Macro Average (%)		
				P	R	F1	P	R	F1
Coarse-Grained									
Baseline	–	CRF	BIO	78.83	40.05	53.12	74.46	32.08	44.37
Fine-tuning	Encoder	MuRIL	BIO	79.84	52.09	63.04	75.46	48.77	58.56
	Enc-Dec	ByT5	BIO	78.96	51.85	62.60	72.53	44.96	54.63
	Decoder	Qwen-2.5-7B	In. Bracketed	79.38	53.37	63.83	58.75	41.75	48.63
	Decoder	Gemma-4-12B	In. Bracketed	78.69	51.91	62.55	73.29	41.89	53.10
Few-Shot	Decoder	Qwen-2.5-7B	Two-Stage	17.85	24.07	20.50	17.27	18.83	15.12
	Decoder	Gemma-4-12B	Two-Stage	48.01	42.81	45.26	31.33	40.28	32.48
	Closed-Source	Gemini 3 Flash	Two-Stage	66.34	76.70	71.14	47.66	69.09	55.05
Fine-Grained									
Baseline	–	CRF	BIO	69.73	33.87	45.59	58.37	23.93	32.80
Fine-tuning	Encoder	MuRIL	BIO	70.16	45.23	55.00	59.38	34.78	41.82
	Enc-Dec	ByT5	BIO	63.21	42.37	50.73	45.07	29.30	34.37
	Decoder	Qwen-2.5-7B	In. Bracketed	67.05	47.96	55.92	52.24	36.53	41.77
	Decoder	Gemma-4-12B	In. Bracketed	69.89	43.38	53.53	49.61	28.07	34.23
Few-Shot	Decoder	Qwen-2.5-7B	Two-Stage	25.38	8.53	12.77	13.24	4.81	6.41
	Decoder	Gemma-4-12B	Two-Stage	38.04	31.19	34.28	22.68	24.00	20.50
	Closed-Source	Gemini-3 Flash	Two-Stage	58.35	65.94	61.91	36.52	48.47	39.80

Table 6: Precision, Recall, and F1 for all models at coarse-grained and fine-grained granularity, grouped by training paradigm (fine-tuning vs. few-shot) and architecture. Bold marks the best score in each column within a granularity.

Model	P	R	F1
CRF	85.86	43.63	57.86
MuRIL	86.45	56.40	68.27
ByT5	85.99	56.48	68.18
Gemini-3 Flash	71.14	82.24	76.29
Qwen-2.5-7B (FT)	82.62	55.55	66.43
Gemma-4-12B (FT)	85.11	56.14	67.66
Qwen-2.5-7B (Fewshot)	36.61	49.38	42.05
Gemma-4-12B (Fewshot)	58.74	52.37	55.37

Table 7: Mention Detection performance across models (coarse-grained setup). (§6).

Model	Coarse (Recall %)		Fine (Recall %)	
	Seen	Unseen	Seen	Unseen
CRF	71.9	8.8	63.7	4.6
MuRIL	80.6	24.0	72.5	18.3
ByT5	81.9	22.4	71.3	13.7
Qwen-FT	81.8	25.1	75.2	21.1
Gemma-FT	77.0	27.2	68.4	18.8
Qwen-FS	30.4	17.8	11.9	5.2
Gemma-FS	59.3	26.5	46.5	16.1
Gemini-3 Flash	86.3	67.2	76.4	55.6

Table 8: Recall (%) for seen vs. unseen entities, based on exact surface-form overlap with training data.

pointwise $\mathcal{V}$ -information for mentions in the test set. To prevent extreme outlier bias, we exclude highly sparse classes when calculating the dataset-level mean $\mathcal{V}$ -information (Table 12). Evaluating these metrics reveals that entity $\mathcal{V}$ -information is negative while context $\mathcal{V}$ -information is positive at both granularities, establishing context as the primary signal.

5 Error Analysis

5.1 Quantitative Analysis

We classify predictions into eight error categories following the taxonomy by (Xie et al., 2023) (see definition in Appendix D), with full distributions reported in Table 11 and summarized (fine-grained) in Figure 3.

Error profiles split cleanly by training paradigm at the coarse-grained level: gradient-trained models (fine-tuned and CRF) are dominated by Omitted Mentions, while few-shot models show comparatively more Completely-O errors. This pattern holds at both granularities (Table 11) and across architectures, suggesting the effect stems from the training paradigm itself.

Wrong-Type errors rise consistently from coarse to fine across nearly every model, showing that fine-grained classification is inherently harder than coarse. Qwen-FS is the sole exception, with Wrong-Type errors dropping from 16.8% to 6.7% possibly because its Omitted Mentions rate jumps sharply (26.5% to 76.3%). We found it driven by its heavy reliance on the UNKNOWN fallback tag we

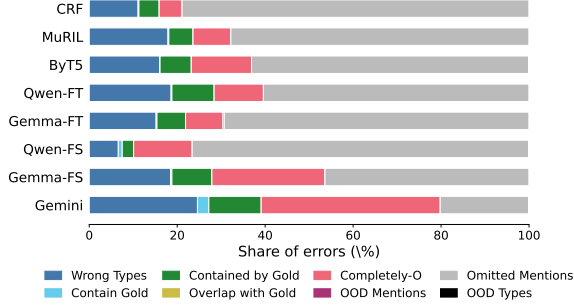


Figure 3: Fine-grained error type distribution by model

used in few-shot classification step (75.1% of fine-grained predictions, vs. 8.0% for Gemma-FS and 2.3% for Gemini). Boundary errors are similarly asymmetric, with under-extension dominating over over-extension across all models, an effect most pronounced in few-shot models.

5.2 Manual Analysis

To understand the error patterns in Table 11, we manually reviewed a sample of model predictions. Wrong Type errors tend to arise mostly in two situations. The first is confusion between fine-grained categories that share the same coarse type. For example, mythical mobile and immobile objects are often mixed up, so weapons get tagged as *alaukika-calanirjīvaḥ* (mythical mobile artifact) instead of *alaukika-acalanirjīvavastu* (mythical immobile artifact). The second situation is category assignment driven by an entity’s relationships rather than the entity itself. Gemini, for instance, tags the goddess Dānu as *devayoniḥ* (semi-divine being) in one context and *devatā* in another, depending on which relation, mother of the Dānavas or daughter of Dakṣa, is most salient in the surrounding text, rather than the entity’s own consistent type. Under-extension errors tend to arise when a model stops reading at the first word boundary instead of continuing to the full entity span. For example, Gemini tags only *baka* instead of the full phrase *baka vadhaḥ*, or dropping *-nandana* from *keśava-nandana* which stems from frequent use of compounds and relational identifiers as mentions in classical Sanskrit texts.

6 Ablation Study

Coarse-level oracle detection: We ablate the few-shot Gemma setup by replacing predicted spans with gold spans and rerunning coarse classification to isolate detection from classification error.

Setting	Micro F1	Macro F1
Coarse, oracle detection	66.1	50.1
Fine, w/ coarse (oracle)	47.5	34.9
Fine, w/o coarse (oracle)	45.7	29.4

Table 9: Oracle-detection ablations for Gemma.

F1 rises from 45.26/32.48 to 66.1/50.1 (micro/macro), showing detection as the bottleneck. Classification with oracle detection exceeds fine-tuned Gemma’s micro F1 (62.55) but trails its macro F1 (53.10 vs. 50.1). MISC shows high recall but low precision (73.4/13.0), acting as a default under uncertainty.

Fine-level oracle detection: We repeat the oracle-detection ablation at the fine-grained level using gold spans, comparing two settings: predicting the fine label given the gold coarse label (w/ coarse) versus predicting it from the gold span alone (w/o coarse). Both (47.5/34.9 and 45.7/29.4 F1; Table 9) trail coarse-oracle despite identical oracle detection, showing granularity, not only detection, drives the difficulty. Sparse classes like *calanirjīvaḥ* and *mānavanirmitaḥ* collapse toward zero F1, likely from too few instances for effective KNN retrieval.

7 Conclusion

We built *Padārtha:Ontology-Grounded Fine-Grained NER Benchmark for Classical Sanskrit*, by grounding the tagset in Nyāya–Vaiśeṣika rather than adapting a schema designed for modern news. The resulting 18 categories sit under 10 hierarchical nodes and collapse to five standard coarse tags, so the data remains usable with existing tools. Labeling 12.6K entries of Sørensen’s *Index* and projecting these through the *Mahānāma* links gives 108,335 mentions over 73,632 verses, with a 5,000-verse test set corrected by hand.

Our benchmarking reveals that while task-specific designs on Sanskrit are now comparable with refined generative models, neither model performs well in the fine-grained context. The V-information analysis points to why: fine categories carry more of their signal in the surrounding context than in the mention itself, which is exactly what a memorizing model cannot exploit.

Limitations

All experiments use machine-segmented text; we do not address the additional challenges posed by

sandhi in Sanskrit, which we leave for future work. Our benchmark is built on a classical epic in verse form, so applicability to prose texts may be limited and would require further investigation, for instance using poetry-to-prose conversion techniques. We do not address the class imbalance present in the dataset, which we leave for future investigation. The source corpus also contains some OCR and automatic segmentation errors; assessing their impact is out of scope for this work, which instead focuses on a standardized classification guideline grounded in the ontology.

Ethics Statement

The annotations in this work are derived from published, copyright-free sources and a publicly available corpus (Sarkar et al., 2025; Sørensen, 1904). All resources utilized have been appropriately cited. The dataset, including annotations, is constructed entirely from existing literary sources; no explicit bias analysis has been performed. The dataset, annotations, and code will be released under a CC-0 license. Annotation was carried out by a doctoral researcher in Sanskrit, who is also an author of this paper, and who assigned fine-grained labels to each of the 12.6K Index entries and manually reviewed and corrected the 5,000-verse test set. Annotation reliability was assessed by comparing the primary annotator’s labels against a postdoctoral Sanskrit grammarian with no prior exposure to the Index of Names, over 1,000 doubly annotated test verses; this review was conducted as a courtesy and was not compensated. The dataset does not contain any personal or sensitive information.

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AI Assistance

AI assistants were used during the writing process to refine textual clarity, grammar, and phrasing. AI assistants were also used during development to assist with debugging code used in the experimental pipeline.

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A Dataset Statistics

Table 10 reports the number of entity mentions per category in the training and test splits, for both coarse- and fine-grained schemas.

B Implementation Details

CRF. We train a linear-chain CRF (sklearn-crfsuite) using L-BFGS optimization ($c_1 = 0.1$, $c_2 = 0.1$, max 100 iterations, all possible transitions enabled), on shallow surface features: token identity, character prefixes/suffixes (1–3 characters), a ± 1 token context window, and sentence-boundary markers.

MuRIL. We fine-tune google/muril-large-cased using the SpanMarkerNER framework (max sequence length 512, entity max length 8), with learning rate $1e-5$, warmup ratio 0.1, batch size 8, up to 10 epochs with early stopping (patience 3), fp16 precision.

Entity Class	Train	Test
Coarse-Grained		
person	72,503	16,427
norp	7,319	2,097
misc	3,577	960
location	2,879	1,919
time	558	96
Total (Coarse)	86,836	21,499
Fine-Grained		
manuṣyaḥ	45,377	6,592
devatā	8,217	2,932
samūhaḥ	7,319	2,097
īśvaraḥ	7,228	4,043
ṛṣiḥ	5,195	1,229
devayoniḥ	5,080	942
śabdaḥ	2,498	634
prākṛtikasthānam	1,721	1,563
alaukikapraṇī	1,305	634
alaukika_acalanirjīvavastu	649	207
alaukikasthānam	628	178
kālaḥ	558	96
janapadaḥ	523	167
acalanirjīvavastu	408	104
jantuh	101	55
alaukikacalanirjīvaḥ	15	5
calanirjīvaḥ	7	10
mānavanirmitaḥ	7	11
Total (Fine)	86,836	21,499

Table 10: Distribution of entity mentions across the training and test sets for both coarse-grained and fine-grained labeling schemas.

ByT5. We fine-tune chronbmm/sanskrit5-multitask for sequence-to-sequence tag generation, using learning rate $3e-4$, batch size 8, 15 epochs, max sequence length 512, linear scheduler with 100 warmup steps.

Qwen2.5-7B Fine-Tuning. We fine-tune Qwen/Qwen2.5-7B-Instruct via LoRA (rank 32, $\alpha = 64$) using the LLaMA-Factory framework⁶, learning rate $2e-5$, effective batch size 16, 3 epochs, bf16 precision, max sequence length 1536, cosine scheduler with warmup ratio 0.05.

Gemma-4-12B Fine-Tuning. We fine-tune google/gemma-4-12b-it via LoRA (rank 32, $\alpha = 64$) using Unsloth⁷ (chosen as unavailability of model in LLaMA-Factory at the time of experiment and the mode for memory efficiency at this model size), learning rate $1e-4$, effective batch size 16, 3 epochs, cosine scheduler with warmup ratio 0.05, bf16 precision, max sequence length 1536, AdamW optimizer, weight decay 0.01.

⁶<https://github.com/hiyouga/LLaMAFactory>

⁷<https://unsloth.ai/>

Qwen2.5-7B and Gemma-4-12B Few-Shot. bf16 precision, temperature 1.0, top- p 0.95.

Computational Budget. All fine-tuning and inference for MuRIL, ByT5, Qwen2.5-7B, and Gemma-4-12B were performed on a single machine with $2 \times$ NVIDIA L40 GPUs (46GB each).

C Prompt Templates

Figure 4 shows the full instruction templates used for supervised fine-tuning, at both fine-grained and coarse-grained levels. Figure 5 shows the two-stage few-shot cascade prompts. Few-shot inference uses the same tagset definitions within a two-stage cascade: a mention-extraction prompt (favoring recall) followed by a classification prompt assigning each detected span a category.

D Error Taxonomy

We categorize sequence tagging discrepancies into eight distinct error types to diagnose typological, boundary, and detection failures:

- **Wrong Type:** The predicted boundary is exact, but the assigned category is incorrect.
- **OOD Type (Out-of-Domain):** The predicted category falls outside the predefined label schema.
- **Contained by Gold (Under-extension):** The prediction only captures a sub-part of the true entity span.
- **Contain Gold (Over-extension):** The prediction completely covers the true entity but incorrectly includes adjacent non-entity text.
- **Overlap with Gold:** A partial match where neither the predicted nor the true span fully contains the other.
- **Completely-O (False Positive):** The predicted span has zero overlap with any ground-truth entity.
- **Omitted Mention (False Negative):** A ground-truth entity is entirely missed by the model.
- **OOD Mention (Hallucinated):** The model predicts an entity using a text string that does not physically exist in the input verse.

Table 11 reports the complete error distribution across all models at both coarse and fine granularity, underlying the summary in Figure 3.

Model	Gran.	Wrong Types	Contain Gold	Contained by Gold	Overlap with Gold	Completely-O	OOD Ment.	Omitted Mentions	OOD Types
CRF	Coarse	5.6%	0.3%	5.1%	0.0%	5.9%	0.0%	83.1%	0.0%
CRF	Fine	11.1%	0.2%	4.6%	0.0%	5.2%	0.0%	78.9%	0.0%
MuRIL	Coarse	8.2%	0.2%	6.6%	0.0%	9.8%	0.0%	75.2%	0.0%
MuRIL	Fine	17.9%	0.2%	5.5%	0.0%	8.6%	0.0%	67.8%	0.0%
ByT5	Coarse	8.6%	0.3%	6.7%	0.0%	10.2%	0.0%	74.2%	0.0%
ByT5	Fine	16.0%	0.1%	7.1%	0.0%	13.7%	0.0%	63.0%	0.0%
Qwen-FT	Coarse	3.8%	0.5%	5.8%	0.1%	14.0%	4.5%	71.3%	0.0%
Qwen-FT	Fine	19.0%	0.2%	9.7%	0.0%	11.3%	0.0%	59.8%	0.0%
Gemma-FT	Coarse	7.8%	0.3%	7.4%	0.0%	10.4%	0.4%	73.5%	0.0%
Gemma-FT	Fine	15.4%	0.2%	6.5%	0.0%	8.2%	0.3%	69.5%	0.0%
Qwen-FS	Coarse	16.8%	1.9%	5.0%	0.0%	49.6%	0.0%	26.5%	0.0%
Qwen-FS	Fine	6.7%	0.9%	2.6%	0.0%	13.5%	0.0%	76.3%	0.0%
Gemma-FS	Coarse	11.4%	0.4%	10.9%	0.0%	32.2%	0.0%	45.1%	0.0%
Gemma-FS	Fine	18.8%	0.3%	9.2%	0.0%	26.1%	0.0%	45.5%	0.0%
Gemini	Coarse	11.6%	3.4%	14.7%	0.1%	51.4%	0.0%	18.8%	0.0%
Gemini	Fine	24.9%	2.7%	12.0%	0.0%	41.4%	0.0%	19.0%	0.0%

Table 11: Error distribution across models and granularities. Percentages are relative to total errors.

D.1 Per-Label Performance Heatmaps

To complement the aggregate micro/macro F1 scores reported in the main text, we provide a per-label breakdown of F1 performance across all eight evaluated models, at both the coarse-grained (5-category) and fine-grained (18-category) levels. Figure 6 shows coarse-grained and fine-grained results; darker green indicates higher F1, darker red indicates lower F1, with the exact value printed inside each cell.

E V-Information

We measure test dataset difficulty using $\mathcal{V}$ -usable information, adapted for NER dataset bias analysis by Ma et al. (2023). Following their method, we decouple entity and context by constructing two datasets from our NER data: an entity-only dataset (using only the entity span) and a context-only dataset (masking the entity with [MASK], retaining the surrounding sentence). We fine-tune MuRIL-large separately on each to predict entity type.

For each test instance (x, y) , pointwise $\mathcal{V}$ -information (PVI) is

$$\text{PVI}(x \rightarrow y) = -\log_2 g[\emptyset](y) + \log_2 g'[x](y),$$

where $g[\emptyset]$ is a null-input baseline estimated from training label frequencies, and $g'[x]$ is the entity-only or context-only model. Entity and Context $\mathcal{V}$ -information are the mean PVI across test instances for each model.

Table 12 reports results at both granularities. Both show negative Entity $\mathcal{V}$ -information and positive Context $\mathcal{V}$ -information, indicating classification relies more on context than entity regularities. This gap is larger at fine granularity, suggesting fine-grained distinctions are more context-dependent.

Granularity	Entity $\mathcal{V}$ -Info	Context $\mathcal{V}$ -Info
Coarse (5 tags)	-0.18	+0.41
Fine-grained (18 tags)	-0.31	+1.19

Table 12: $\mathcal{V}$ -information for entity vs. context classifiers.

F Minority-Sense Evaluation

Model	Correct	Wrong	N.Det.
CRF	8.3%	25.7%	66.0%
Gemini-3 Flash	59.5%	27.1%	13.3%
Qwen-FT	31.8%	32.1%	36.0%
Qwen-FS	4.8%	6.7%	88.5%
MuRIL	24.2%	32.6%	43.2%
Gemma-FT	25.3%	27.4%	47.4%
Gemma-FS	17.7%	28.9%	53.4%

Table 13: Fine-grained accuracy on minority-sense homonyms. N.Det. = Not detected.

G Related Works

The current state of Sanskrit Named Entity Recognition (NER) is marked by a transition toward large-scale data creation, yet a critical gap remains. Most existing resources are either coarse grained or classified as silver standard datasets derived through automated methods, and comprehensive benchmarking is largely absent. Early literary efforts, such as the pre-annotation based approach using the Śrīmad Bhāgavatam (Sujoy et al., 2023), rely on a corpus of 18,000 verses and identify entities through a semi-automated heuristic that compares transliterated Sanskrit with English translations. This work does not employ traditional neural NER models, instead relying on string matching algorithms such as Jaro Winkler similarity to generate entity suggestions for annotators. The dataset is limited to coarse grained tags, specifically Person, Location, and Miscellaneous, and excludes organizational entities due to their absence in spiritual texts.

Similarly, the Mahānāma project (Sarkar et al., 2025), derived from the Mahābhārata, identifies 109,000 mentions but focuses primarily on Entity Discovery and Linking and coreference resolution. Its experiments evaluate coreference and entity linking models rather than dedicated NER classification systems. It follows a similar coarse grained schema and lacks standardized annotation guidelines for Sanskrit, relying on an expert curated name index to define entities.

To address the scarcity of training data, recent frameworks have introduced large scale silver standard corpora, though these often lack the linguistic precision of manually annotated gold data. The Naamah corpus (P and Kulkarni, 2026) provides a benchmark with 102,942 sentences and 127,397 total entities, including 90,452 Person, 22,290 Location, and 14,655 Organization tags. This dataset is used to benchmark transformer architectures, comparing XLM RoBERTa Base with the parameter efficient IndicBERTv2. However, it is synthetically generated using the Sarvam M model via DBpedia seeding and may inherit biases from prompting templates.

In the multilingual setting, SampurNER (Kaushik and Anand, 2026) and TAFSIL (Kaushik et al., 2025) extend fine grained NER to Sanskrit within broader Indian language initiatives, but both remain silver standard due to their methodologies. SampurNER employs the

Entity Anchored Machine Translation framework to translate the English FewNERD dataset into Sanskrit, yielding 152,269 entities and evaluating models such as mBERT and IndicBERTv2. TAFSIL uses distant supervision by linking Wikipedia with Wikidata to construct datasets across multiple taxonomies, and evaluates noise aware models including LITE and DECENT with encoders such as MuRIL, XLM RoBERTa, and mDeBERTa.

Despite advances in scale, granularity, and model evaluation, a major gap remains in the availability of a domain native, gold standard Sanskrit dataset that does not rely on translation or synthetic generation, along with the absence of systematic and comparable benchmarking across NER approaches.

Fine-Grained Fine-Tuning Instruction

You are an expert Sanskrit linguist performing Named Entity Recognition (NER) on the Mahābhārata. Identify ALL named entities in the Sanskrit sentence. Rewrite the sentence with inline tags using the exact format: [Entity Text | fine_tag].

Entity Types: Use ONLY the 18 tags below.

Person / Beings: (1) *manuṣyaḥ* – mortal humans (Arjuna, Duryodhana); (2) *jantuḥ* – ordinary named animals (Sugrīva, Aśvatthāman); (3) *alaukikapraṇī* – mythical/divine creatures (Ananta, Garuḍa); (4) *īśvaraḥ* – Trimūrti and avatars (Brahmā, Viṣṇu, Śiva, Kṛṣṇa); (5) *devatā* – standard gods (Indra, Bhūmi, Vāyu); (6) *ṛṣiḥ* – sages/ascetics (Vasiṣṭha, Viśvāmitra); (7) *devayoniḥ* – semi-divine/demons (Kubera, Ghaṭotkaca).

Locations: (8) *prākṛtikasthānam* – natural places (Daṇḍakāraṇya, Gaṅgā); (9) *alaukikasthānam* – mythical world-s/heavens/hells (Gandharvaloka, Kālasāhvaya); (10) *janapadaḥ* – geo-political areas (Aṅga, Avantī, Hastināpura); (11) *mānavanirmitaḥ* – man-made structures (Lākṣāgrham, Agastyāśrama).

Artifacts & Texts: (12) *calanirjīvaḥ* – mundane mobile artifacts (Jaitra); (13) *alaukikacalanirjīvaḥ* – mythical mobile artifacts (Puṣpaka); (14) *acalanirjīvavastu* – mundane stationary objects (Asamprṛtha); (15) *alaukika_acalanirjīvavastu* – mythical/celestial stationary objects (Bhārgavāstra, Sudarśana); (16) *śabdaḥ* – named texts/mantras (Yajus, Rāmāyaṇa).

Time: (17) *kālāḥ* – specific units of time/eras (Bhādrapada, Ardhamaśa).

Groups: (18) *samūhaḥ* – specific named groups/clans (Ābhīra, Amarāḥ).

Indirect Entity Spans. Standard proper nouns are always tagged; additionally tag the entire word/phrase when an entity is identified indirectly: (1) *Derived Names* – grammatically created from a name (Brāhmī); (2) *Relational Identifiers* – identifies via relationship (Kuntīputra, Kāśīpati); (3) *Epithets* – descriptive titles used as names (Vṛkodara); **Critical Rules.** (1) Format strictly as [Entity | tag], no spaces around the pipe. (2) Tag specifics only – never generic nouns, unless acting as a proper-name substitute for a specific character (e.g., *ācārya* for Droṇa). (3) No nested tags – if a compound acts as a single identifier, tag the entire compound once. (4) Maintain exact spelling, sandhi, and segmentation outside brackets.

Coarse-Grained Fine-Tuning Instruction

You are an expert Sanskrit linguist performing Named Entity Recognition (NER) on the Mahābhārata. Identify ALL named entities in the Sanskrit sentence. Rewrite the sentence with inline tags using the exact format: [Entity Text | LABEL].

Entity Types: Use ONLY the 5 broad tags below. (1) **PER** (Person/Being) – all sentient beings, including gods, sages, humans, specific animals, and mythical creatures (Brahmā, Kṛṣṇa, Arjuna, Vasiṣṭha, Sugrīva, Ananta); (2) **LOC** (Location) – all places, including natural features, mythical realms, kingdoms, and man-made structures (Gaṅgā, Gandharvaloka, Hastināpura, Lākṣāgrham); (3) **MISC** (Miscellaneous) – inanimate artifacts, vehicles, divine weapons, and texts (Puṣpaka, Bhārgavāstra, Sudarśana, Rāmāyaṇa); (4) **TIME** – specific periods, eras, or units of time (Kṛtayuga, Bhādrapada); (5) **NORP** (Groups) – specific named collectives, lineages, tribes, or religious groups (Ābhīra, Amarāḥ).

Indirect Entity Spans. Standard proper nouns are always tagged; additionally tag the entire word/phrase when an entity is identified indirectly: (1) *Derived Names* – grammatically created from a name (Brāhmī); (2) *Relational Identifiers* – identifies via relationship (Kuntīputra, Kāśīpati); (3) *Epithets* – descriptive titles used as names (Vṛkodara);

Critical Rules. (1) Format strictly as [Entity | tag], no spaces around the pipe. (2) Tag specifics only – never generic nouns, unless acting as a proper-name substitute for a specific character (e.g., *ācārya* for Droṇa). (3) No nested tags – if a compound acts as a single identifier, tag the entire compound once. (4) Maintain exact spelling, sandhi, and segmentation outside brackets.

Figure 4: Fine-tuning instruction templates for fine-grained (top) and coarse-grained (bottom) Sanskrit NER.

Stage 1: Mention Extraction Prompt (Few-Shot)

You are an expert Sanskrit linguist performing Named Entity Recognition (NER) on the Mahābhārata.

Task Requirements: (1) Surround entity spans using @@ and ##. (2) Surround the output sentence with \$\$ and \$\$\$. (3) Do NOT change, remove, or reorder any tokens. (4) Identify ALL possible named entities. Do NOT miss any entity. (5) When in doubt, mark it as an entity. Prefer higher recall over precision.

Entity Types: (1) **PER** (Person/Being) – all sentient beings, including gods, sages, humans, specific animals, and mythical creatures (Brahmā, Kṛṣṇa, Arjuna, Vasiṣṭha, Sugrīva, Ananta); (2) **LOC** (Location) – all places, including natural features, mythical realms, kingdoms, and man-made structures (Gaṅgā, Gandharvaloka, Hastināpura, Lākṣāgrhaṁ); (3) **MISC** (Miscellaneous) – inanimate artifacts, vehicles, divine weapons, and texts (Puṣpaka, Bhārgavāstra, Sudarśana, Rāmāyaṇa); (4) **TIME** – specific periods, eras, or units of time (Kṛtayuga, Bhādrapada); (5) **NORP** (Groups) – specific named collectives, lineages, tribes, or religious groups (Ābhīra, Amarāḥ).

Indirect Entity Spans. Standard proper nouns are always tagged; additionally tag the entire word/phrase when an entity is identified indirectly: (1) *Derived Names* – grammatically created from a name (Brāhmī); (2) *Relational Identifiers* – identifies via relationship (Kuntīputra, Kāśipati); (3) *Epithets* – descriptive titles used as names (Vṛkodara);

Critical Rules. (1) Tag specifics only – never generic nouns, unless acting as a proper-name substitute for a specific character (e.g., *ācārya* for Droṇa). (2) No nested tags – if a compound acts as a single identifier, tag the entire compound once. (3) Maintain exact spelling, sandhi, and segmentation outside brackets.

IMPORTANT: Output ONLY the tagged sentence wrapped in \$\$. . . \$. Nothing else.

Stage 2: Classification Prompt (Few-Shot)

You are an expert Sanskrit linguist performing Named Entity Recognition (NER) on the Mahābhārata. Classify the detected entities into the following labels:

Entity Types: (1) **PER** (Person/Being) – all sentient beings, including gods, sages, humans, specific animals, and mythical creatures (Brahmā, Kṛṣṇa, Arjuna, Vasiṣṭha, Sugrīva, Ananta); (2) **LOC** (Location) – all places, including natural features, mythical realms, kingdoms, and man-made structures (Gaṅgā, Gandharvaloka, Hastināpura, Lākṣāgrhaṁ); (3) **MISC** (Miscellaneous) – inanimate artifacts, vehicles, divine weapons, and texts (Puṣpaka, Bhārgavāstra, Sudarśana, Rāmāyaṇa); (4) **TIME** – specific periods, eras, or units of time (Kṛtayuga, Bhādrapada); (5) **NORP** (Groups) – specific named collectives, lineages, tribes, or religious groups (Ābhīra, Amarāḥ); (6) **UNKNOWN** – use this label if the span is clearly not a named entity.

Output Format: Return ONLY a JSON list of {entity_type}, using k-nearest-neighbor-retrieved examples as in-context demonstrations. No explanations or additional text.

Figure 5: Two-stage few-shot pipeline prompts: mention extraction (Stage 1) followed by classification (Stage 2).

